Subject Description Form

TBC

Subject Code	SD5008
Subject Title	Research and Analysis for Design
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	Research and Analysis for Design is a compulsory common subject that introduces students to design research. Students will critically question research frameworks, methodologies to advance design enquiries, and conceptual foundations. The course introduces advanced design processes, equipping students with skills to identify, translate, and manage theoretical, material, and technological variables. Commencing from single questions, the course encompasses broad-based investigations, to facilitate students to acquire methodologies to test proposed material criticisms against theoretical shifts, demonstrating research outcomes, and possible bodies of work. The main course components are: (i) To nurture the skills needed to establish research frameworks that link research, practice, and design thinking. (ii) To create an awareness of, as well as an appreciation of, the value of research, as a meaningful contribution for design practices to strategize environments, products, and service systems. (iii) To prepare students to conduct independent and impactful research. (iv) To facilitate the planning and execution of research projects, mechanising phases of analysis, data synthesis, and research findings. (v) To develop a body of research applicable to master's level design research.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional skills</u> a. Discuss the role of research in design practice b. Justify the choice of appropriate research methods and their suitability to address a design issue or solve a design problem c. Justify the choice of appropriate research methods and data collection d. Utilise appropriate methods and tools to manage and analyse data e. Synthesise and present research findings using appropriate (re)presentation formats f. Understand, at a foundational level, design research as a professional specialisation <u>Transferable skills</u> g. Communicate ideas effectively in written, oral, and visual formats h. Collaborate with peers, faculty, and stakeholders i. Continuously develop critical assessment of data in design practices

Subject Synopsis/ Indicative Syllabus

Students will be introduced to:

- A variety of approaches to research: research as process, methodologies, outcomes, and critical positioning
- Then planning and execution of research processes
- Approaches to assess the interplay of theory, research, and impact
- Contemporary research methods
- Data management and analysis: tools and techniques
- Data synthesis and (re)-presentation

Teaching/Learning Methodology

Activity	Purpose
Lectures	To introduce students to theories and principles related to design research and related theories and concepts.
Workshops	Workshops consist of hands-on tasks which allow students to experience specific aspects of developing a publishable text, concepts, and methods deepening their understanding of the lecture material.
Guided Study	To guide students on the development of their progress, discuss concepts and knowledge of general relevance to support students' writing individually and in small groups. And to develop students' ability to critique others' work, communicate in a constructive and meaningful manner, and be able to receive feedback from others.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed								
		a	b	c	d	e	f	g	h	i
Research Report	40%		✓	✓	✓	✓	✓	✓	✓	✓
Presentations	30%		✓	✓	✓	✓	✓	✓		
Class Participation & Peer Assessment	15%	✓	✓	✓				✓	✓	✓
In-class Individual Written Assessment	15%									
Total	**100%**									

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Research Report	Reports help students to learn academic writing skills, including formulating an argument; presenting evidence; integrating material from sources; and referencing appropriately. Students will report on field research, analyse data, synthesise findings, develop framing of problems/opportunities and generate concepts. Students will also synthesize and evaluate relevant theoretical ideas and concepts and develop research skills.

	Presentations	Presentations help students master oral communication and persuasive skills. They learn how to organise arguments using supporting evidence, select relevant material and engage critically with ideas while developing understanding and confidence. Not only do presentations promote idea sharing in the classroom they also allow for immediate feedback from the lecturer and peers.
	Class Participation & Peer Assessment	Participation marks engage students with course learning and develop their ability to communicate and discuss ideas. These improve levels of participation in the class and provide immediate feedback on the learning. Peer assessment develops an ability to critique others' work, communicate in a constructive and meaningful manner, and be able to receive feedback from others.
	In-class Individual Written Assessment	The In-class Individual Written Assessment provides a controlled environment to evaluate students' independent understanding and mastery of core concepts covered throughout the subject. It tests their ability to synthesize theoretical knowledge, critically analyze research methodologies, and articulate coherent arguments under time constraints without reliance on collaborative support or secondary aids. This ensures academic integrity and reinforces individual accountability for key learning outcomes.

Student Study Effort Expected	Class contact:	
	▪ Lecture	5 Hrs.
	▪ Seminar	4 Hrs.
	▪ Workshop	5 Hrs.
	▪ Guided Study	13 Hrs.
	▪ Project	7 Hrs.
	▪ Presentation	5 Hrs.
	Other student study effort:	
	▪ Self-study	66 Hrs.
	Total student study effort	**105 Hrs.**

Reading List and References

- Hanington, Bruce, and Bella Martin. *Universal Methods of Design: 100 Ways to Research Complex Problems, Develop Innovative Ideas, and Design Effective Solutions.* Rockport Publishers, 2012
- de Jong, Taeke M., and D. J. M. Van Der Voordt, eds. *Ways to study and research: urban, architectural, and technical design*. IOS Press, 2002.
- Brenner, Neil. *New urban spaces: Urban theory and the scale question*. Oxford University Press, 2019.
- Karlsen, James, and Miren Larrea. *Territorial development and action research: innovation through dialogue*. Routledge, 2016.